Conformational Analysis

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0.1 Rotamers

Rotamers, short for rotational isomers, are a specific type of conformational isomer. They are different spatial arrangements of the same molecule that arise purely from rotation around a single (σ) bond.

As we have repeatedly seen throughout the *Introductory Chemistry* notes, matter arranges itself in a way that minimizes its PE. Molecule arrangements are no different.

Let's take a look at different conformational isomers of ethane through a **Newman projection**.

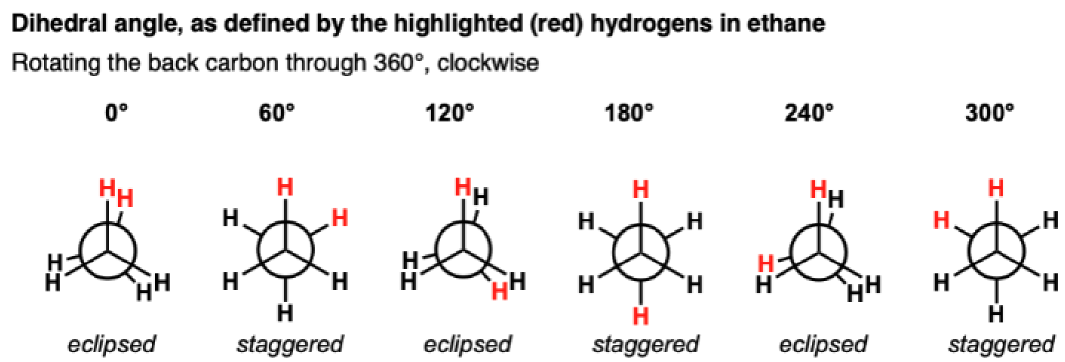

Figure 1: caption

As shown above, there are two different types of positions ethane can take when rotated about its C-C σ bond: **eclipsed** and **staggered**. Of the two, the most stable rotamer is the staggered one because the e^- e^- repulsions between the C-H σ bonds are minimized.

On the flipside, the eclipsed conformation is the least stable because the e^- e^- repulsions between the C-H σ bonds are maximized.

The PE diagram throughout a full rotation around the C-C σ bond looks like the following.

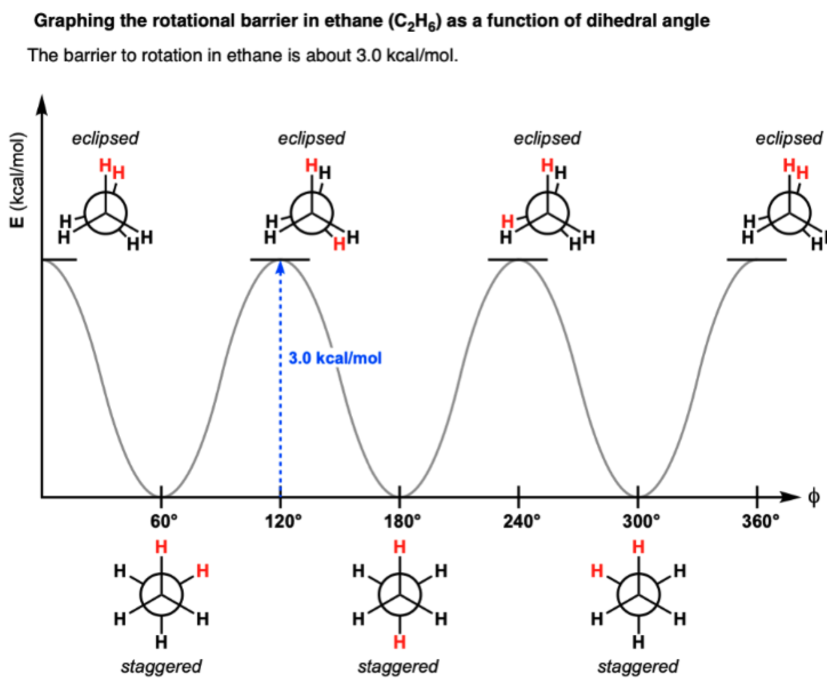

Figure 2: caption

Here, the $3.0 kcal\ mol^{-1}$ would represent the activation energy (E_a) to carry out a rotation from a staggered to an eclipsed conformation. Since the different staggered positions of ethane are exactly identical, there is no net change in the energy of the system after a rotation occurs (neither endothermic nor exothermic). We also see the same pattern in the case of propane.

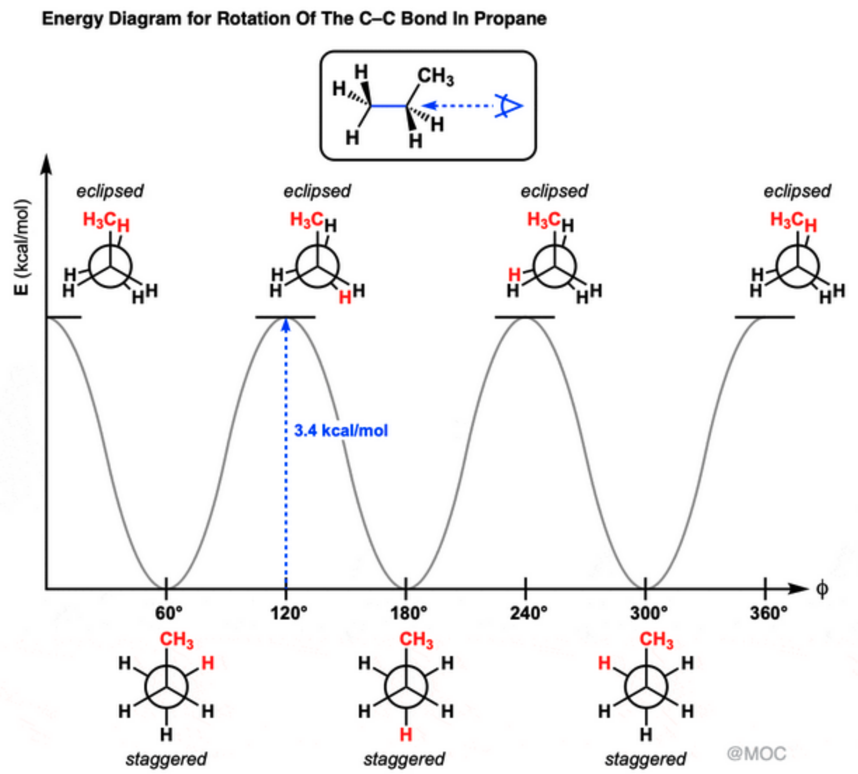

Figure 3: caption

The only difference we see is the E_a increasing to 3.4 $kcal\ mol^{-1}$ for propane. This is because there is greater e^- density within the **methyl of propane** compared to the **hydrogen of ethane**. Thus the e^- e^- repulsion forces are greater.

Let's take a look at one more example: butane.

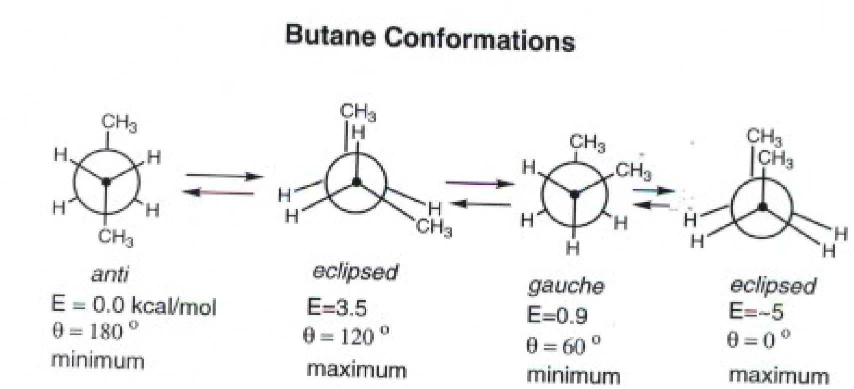

Figure 4: caption

Using the same logic we learned from the greater e^- density consequences in propane compared to ethane, we can see that the anti staggered conformation would be the most stable becasue the methyl groups are placed farthest apart. Second most stable would be the gauche staggered conformation. The fully eclipsed conformation (θ = 0°) would be the least stable because the high e^- density methyls are placed exactly overlapping on one another, whereas the half eclipsed conformation (θ = 120°) would be more stable but still unstable in the grand scheme of the molecule's PE levels.

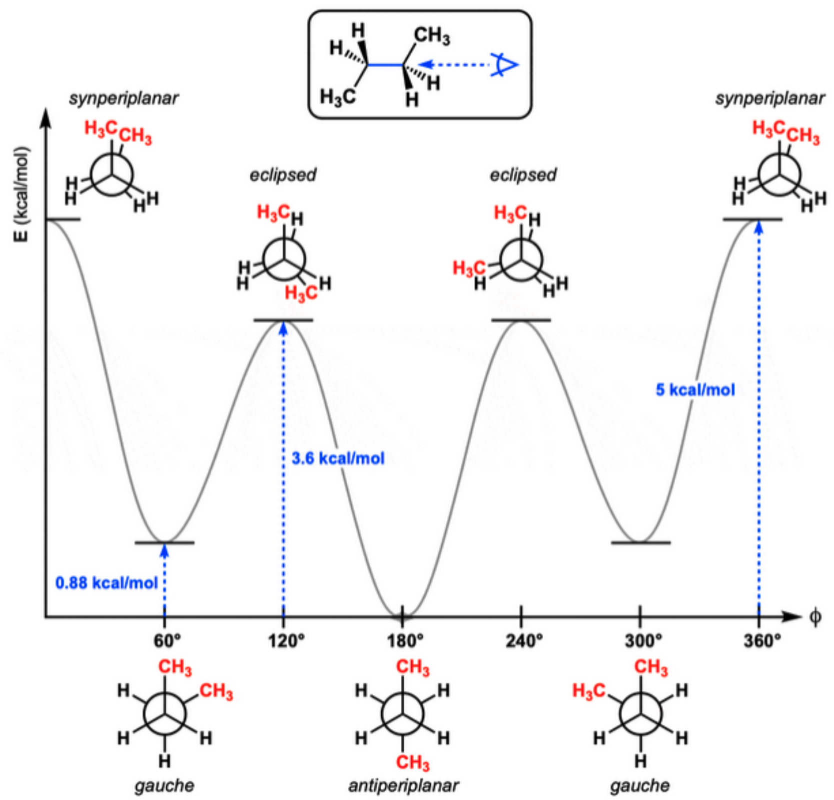

Figure 5: caption

Core Concept 0.1: placeholder

Stereoisomers are different spatial arrangements of the same molecule (same connectivity) that arise from differences in 3D arrangements. Two types of stereoisomers exist: **conformational** and **configurational** isomers.

Conformational isomers are the result of bond rotations that happen freely at room temperature. *Rotamers are conformational isomers.*

Configurational isomers are permanent, meaning they can only be change by the breaking and formation of new bonds. They refer to the different spatial arrangements like cis/trans and R/S.

0.2 Cycloalkane & Ring Strain

Cycloalkanes experience ring strain (**heightened PE**) due to the following factors.

(1) **Bond angle strain** (*especially for smaller rings*)

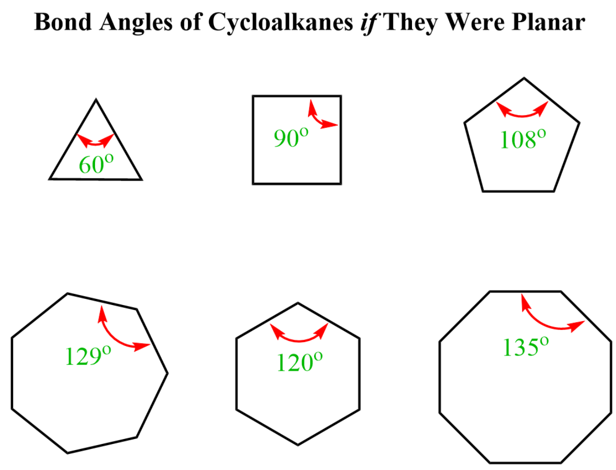

Figure 6: caption

(2) **Eclipsing strain** (*as we have discussed in 1.2.1 Rotamers*)

(3) **Transannular strain** (*the steric hinderence that occurs from atoms "bumping into each other" across the empty space in the middle of the ring*)

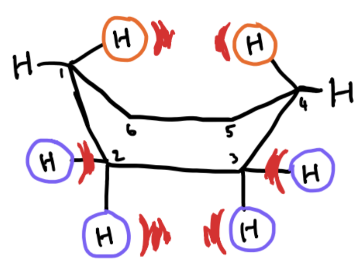

Figure 7: caption

Thus, we see a variety of 3D arrangements across cycloalkanes that differ from the flat ring orientation that we intuitively expect.

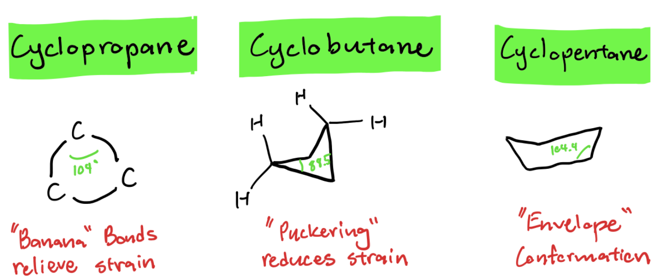

Figure 8: caption

Specifically, we will be focusing on the different conformational arrangements of **cyclohexane**, which most stably exists in a chair shape conformation.

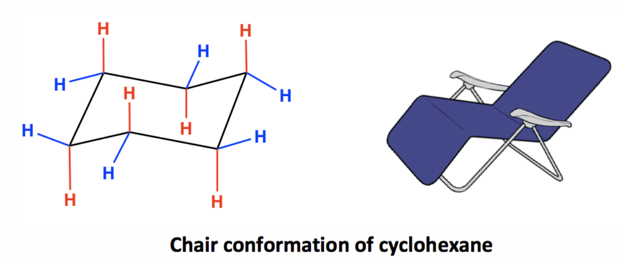

Figure 9: caption

MINIMIZING ECLIPSING STRAIN: One reason why this conformation is the most stable is because the of the staggered rotamer conformation when viewing the Newman projection along any of the C-C bonds as we have learned in **1.2.1 Rotamers**, eliminating the eclipsing strain experienced *if* the cyclohexane took a planar conformation.

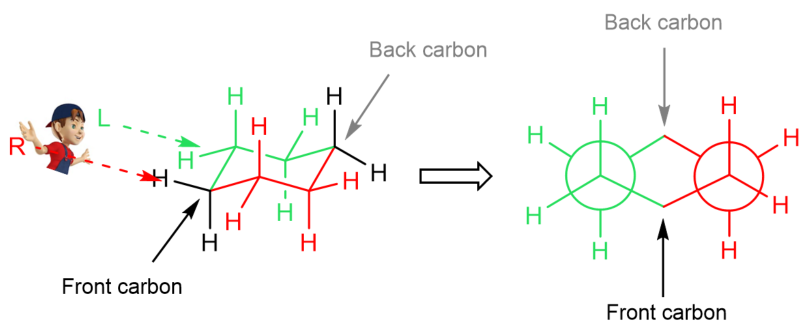

Figure 10: caption

MINIMIZING TRANSANNULAR STRAIN: Another reason why the chair conformation is favored is because the conformation minimizes steric hinderence across the ring. *Eclipsing strain is also increased in this conformation.* Take a look at the **boat conformation** of cyclohexane for instance.

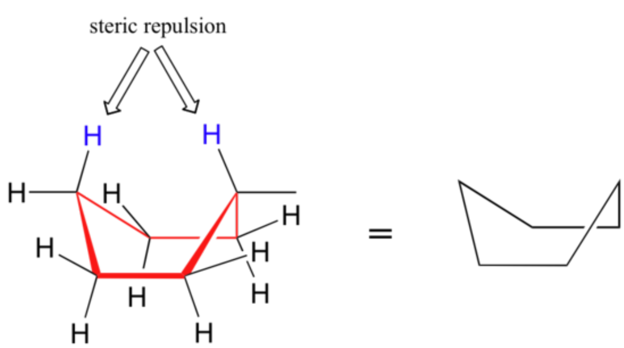

Figure 11: caption

The boat arrangement is actually a transition state in the dynamics of cyclohexane movement. Thus it does not exist readily in the environment. Instead, a slightly adjusted, more stable version is the **twist (skew) boat conformation**.

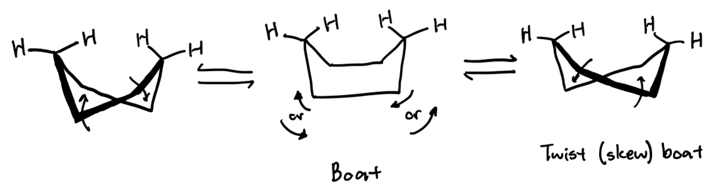

Figure 12: caption

Here is a PE graph across different conformations of cyclohexane.

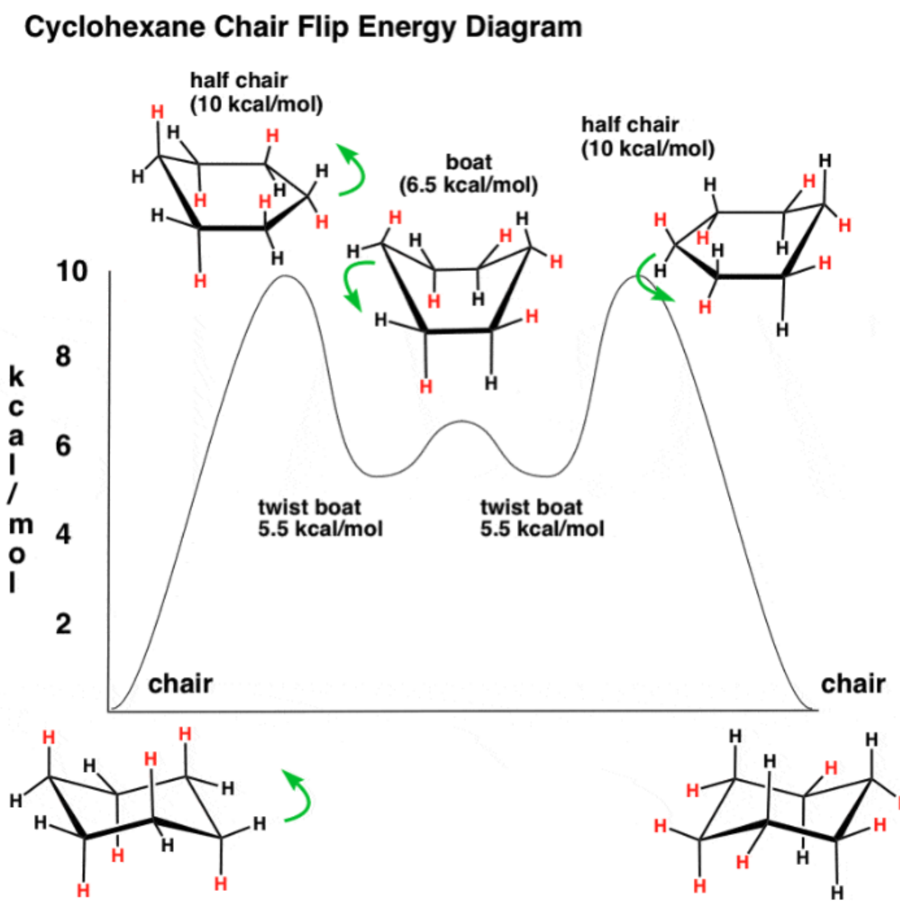

Figure 13: caption

As you can see, there is another high energy (*for the same reasons we have discussed*) transition state, the **half chair conformation**.

While nonsubstituted cyclohexane shows indifference to the two different chair conformations in regards to PE levels, it's important to keep in mind that monosubstituted cyclohexanes (such as methylcyclohexane) do show a preference between the two chair conformation due to differing levels of transannular strain.

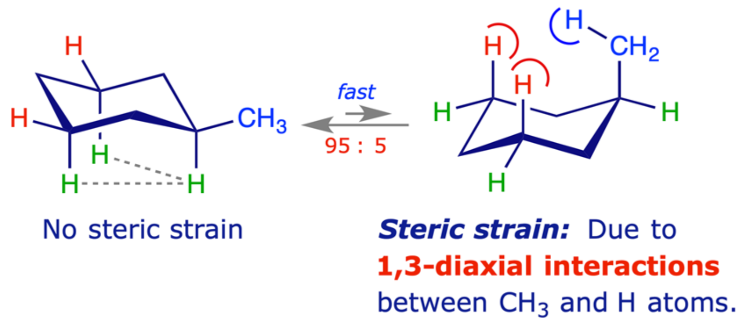

Figure 14: caption

In other words, the **equitorial** methyl chair conformation is preferred over the **axial** methyl chair conformation due to the lower transannular strain.

Core Concept 0.2: placeholder

Axial bonds refer to bonds that point straight up or straight down, perpendicular to the "plane" of the ring. On the other hand, **Equatorial** bonds are those that point outward toward the "equator" or perimeter of the ring.

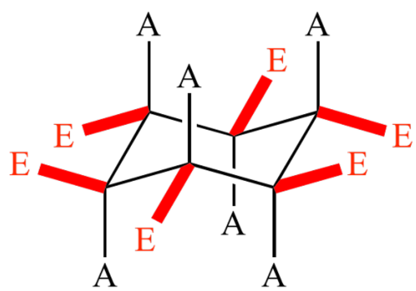

Figure 15: caption